

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

Student Name: M.Siva ganesh

Roll No: A24126552093

Date: 30-08-2026

1. TITLE

Develop a Styled Webpage Using HTML5 Canvas and CSS3

2. AIM

To design and develop a styled webpage using HTML5 and CSS3, featuring a dark-themed layout with HTML5 Canvas graphics, gradient backgrounds, box and text shadows, rounded corners, and a Flexbox-based responsive card layout with interactive hover effects on navigation links, buttons, and content cards.

3. OBJECTIVE

The objectives of this experiment are:

- To understand and use the HTML5 <canvas> element for rendering graphics.
- To draw basic shapes — a filled rectangle, a filled and stroked circle (arc), and a straight line — using the Canvas 2D drawing API.
- To render text on a canvas using the fillText() method with a custom font.
- To apply CSS3 linear-gradient() backgrounds to the header and to each content card.
- To apply box-shadow and text-shadow for depth and visual emphasis on a dark colour theme.
- To use border-radius for rounded corners on containers, the canvas, cards, and buttons.
- To lay out the technology cards using CSS3 Flexbox with wrapping for responsiveness.
- To apply :hover states on navigation links, the intro button, and card buttons for interactivity.
- To structure the page using semantic HTML5 elements: header, nav, section, and footer.

4. THEORY

HTML5 Canvas is a rectangular drawing surface exposed through the <canvas> element. JavaScript accesses a 2D rendering context via getContext('2d') and issues drawing commands — fillRect() for the rectangle, arc() with fill()/stroke() for the circle, moveTo()/lineTo() with stroke() for the line, and fillText() for the name — to render graphics dynamically in the browser.

CSS3 Gradients — linear-gradient() blends two or more colours along a defined direction, producing the smooth dark tonal transitions used on the header background and on each card background, without needing image files.

Box Shadow & Text Shadow — box-shadow adds a layered shadow around an element's box, used here on the intro section, canvas container, canvas, and cards; text-shadow adds a shadow behind text, used on the header heading, to create depth against the dark background.

Border Radius — border-radius rounds the corners of an element's box, used on the intro box, canvas, cards, and buttons for a softer, modern look.

Flexbox — the .card-container uses display: flex with justify-content: center and flex-wrap: wrap to arrange the HTML5, CSS3, and JavaScript cards in a centred row that re-flows onto new lines on narrower screens.

:hover Pseudo-class — the :hover pseudo-class changes the colour and background of navigation links and slightly enlarges buttons when the mouse pointer is over them, giving immediate visual feedback on interactive elements.

5. ALGORITHM / PROCEDURE

- Create an HTML5 file named My_Styled_Webpage.html.
- Add the <!DOCTYPE html> declaration and set the language attribute on <html>.
- Inside <head>, include the charset meta tag, viewport meta tag, page title, and an internal <style> block.
- In the <style> block, reset the <body> margin/padding and set a dark background-color (#121212) with a light text colour (#e0e0e0).
- Build the <header> with a linear-gradient background, centred text, and a text-shadow heading.
- Build the <nav> with inline anchor links — Home, Canvas, Technologies, Contact — and a :hover style that changes their colour and background.
- Create an intro <section id="home"> styled as a rounded, shadowed dark box containing a welcome heading, a paragraph, and an "Explore More" button.
- Create a canvas <section id="canvas"> with a rounded, shadowed dark container holding a <canvas> element.
- Write a <script> block to get the 2D context and draw: one filled rectangle, one filled and stroked circle (arc), one line, and the student's name using fillText().
- Create a cards <section id="cards"> using a flexbox .card-container holding three .card <div> elements for HTML5, CSS3, and JavaScript.
- Style each .card with a gradient background, border, border-radius, and box-shadow, and add a styled button inside each card.
- Add a <footer id="contact"> with a dark background, a copyright line, and a task label.
- Open the file in a web browser and verify the canvas drawing and all CSS3 effects.
- Record expected vs. actual output for each test case.

6. PROGRAM

My_Styled_Webpage.html

```
<!DOCTYPE html>
<html lang="en">

<head>
```

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Styled Webpage</title>

<style>
  /* Page styling */
  body {
    margin: 0;
    padding: 0;
    font-family: Arial, sans-serif;
    background-color: #121212;
    color: #e0e0e0;
  }

  /* Header */
  header {
    background: linear-gradient(to right, #000000, #1f1f1f);
    color: #ffffff;
    text-align: center;
    padding: 20px;
    text-shadow: 2px 2px 3px #000000;
  }

  header h1 {
    font-size: 32px;
    font-family: "Trebuchet MS", sans-serif;
  }

  /* Navigation */
  nav {
    background-color: #0a0a0a;
    padding: 12px;
    text-align: center;
  }

  nav a {
    color: #bbbbbb;
    text-decoration: none;
    margin: 0 15px;
    padding: 7px;
    font-size: 16px;
  }

  nav a:hover {
    color: #ffffff;
    background-color: #2a2a2a;
    border-radius: 5px;
  }

  /* Introduction */
  .intro {
    text-align: center;
    background-color: #1e1e1e;
    margin: 20px;
    padding: 20px;
    border: 2px solid #333333;
    border-radius: 10px;
    box-shadow: 3px 3px 8px #000000;
  }
}
```

```

/* Buttons */
button {
    margin-top: 10px;
    padding: 9px 16px;
    border: 1px solid #333333;
    border-radius: 6px;
    background-color: #2b2b2b;
    color: #ffffff;
    cursor: pointer;
}

button:hover {
    background-color: #3d3d3d;
    transform: scale(1.05);
}

/* Canvas section */
.canvas-section {
    text-align: center;
    background-color: #1a1a1a;
    margin: 20px;
    padding: 20px;
    border: 2px solid #2d2d2d;
    border-radius: 10px;
}

canvas {
    background-color: #242424;
    border: 2px solid #404040;
    outline: 2px dashed #555555;
    border-radius: 6px;
    box-shadow: 3px 3px 8px #000000;
}

/* Cards */
.card-container {
    display: flex;
    justify-content: center;
    gap: 20px;
    padding: 30px;
    flex-wrap: wrap;
}

.card {
    background: linear-gradient(to bottom, #252525, #181818);
    border: 2px solid #333333;
    border-radius: 10px;
    padding: 20px;
    margin: 10px;
    width: 220px;
    box-shadow: 3px 3px 8px #000000;
    text-align: center;
}

.card h3 {
    font-family: Georgia, serif;
    font-size: 23px;
    color: #ffffff;
}

```

```

        /* Footer */
        footer {
            background-color: #0a0a0a;
            color: #888888;
            text-align: center;
            padding: 15px;
            margin-top: 20px;
        }
    </style>
</head>

<body>

    <!-- Main heading -->
    <header>
        <h1>My Styled Webpage</h1>
        <p>HTML5 Graphics and CSS3 Web Styling</p>
    </header>

    <!-- Navigation bar -->
    <nav>
        <a href="#home">Home</a>
        <a href="#canvas">Canvas</a>
        <a href="#cards">Technologies</a>
        <a href="#contact">Contact</a>
    </nav>

    <!-- Introduction -->
    <section id="home" class="intro">
        <h2>Welcome</h2>
        <p>This webpage demonstrates HTML5 Canvas and CSS3 styling.</p>
        <button>Explore More</button>
    </section>

    <!-- Part A: HTML5 Canvas -->
    <section id="canvas" class="canvas-section">
        <h2>Part A - HTML5 Canvas</h2>
        <p>The canvas contains a rectangle, circle, line and my name.</p>

        <canvas id="myCanvas" width="500" height="280"></canvas>

        <script>
            // Select canvas
            const canvas = document.getElementById("myCanvas");
            const ctx = canvas.getContext("2d");

            // Draw rectangle
            ctx.fillStyle = "#1e3a8a";
            ctx.fillRect(40, 40, 130, 80);

            // Draw circle
            ctx.beginPath();
            ctx.arc(350, 80, 45, 0, 2 * Math.PI);
            ctx.fillStyle = "#064e3b";
            ctx.fill();
            ctx.strokeStyle = "#047857";
            ctx.stroke();

            // Draw line
            ctx.beginPath();

```

```

        ctx.moveTo(50, 170);
        ctx.lineTo(450, 170);
        ctx.strokeStyle = "#e2e8f0";
        ctx.lineWidth = 3;
        ctx.stroke();

        // Display name
        ctx.font = "26px Arial";
        ctx.fillStyle = "#ffffff";
        ctx.fillText("ganesh", 220, 230);
    </script>
</section>

<!-- Part B: CSS3 Styling -->
<section id="cards">
    <h2 style="text-align:center; color: #ffffff;">Part B - CSS3 Styling</h2>

    <div class="card-container">

        <div class="card">
            <h3>HTML5</h3>
            <p>HTML5 provides the structure and content of a webpage.</p>
            <button>Learn HTML</button>
        </div>

        <div class="card">
            <h3>CSS3</h3>
            <p>CSS3 is used to style webpages with colors, fonts and layouts.</p>
            <button>Learn CSS</button>
        </div>

        <div class="card">
            <h3>JavaScript</h3>
            <p>JavaScript adds interaction and functionality to webpages.</p>
            <button>Learn JS</button>
        </div>

    </div>
</section>

<!-- Footer -->
<footer id="contact">
    <p>&copy; 2026 My Styled Webpage</p>
    <p>HTML5 & CSS3 Observation Task</p>
</footer>

</body>

</html>

```

7. CODE EXPLANATION

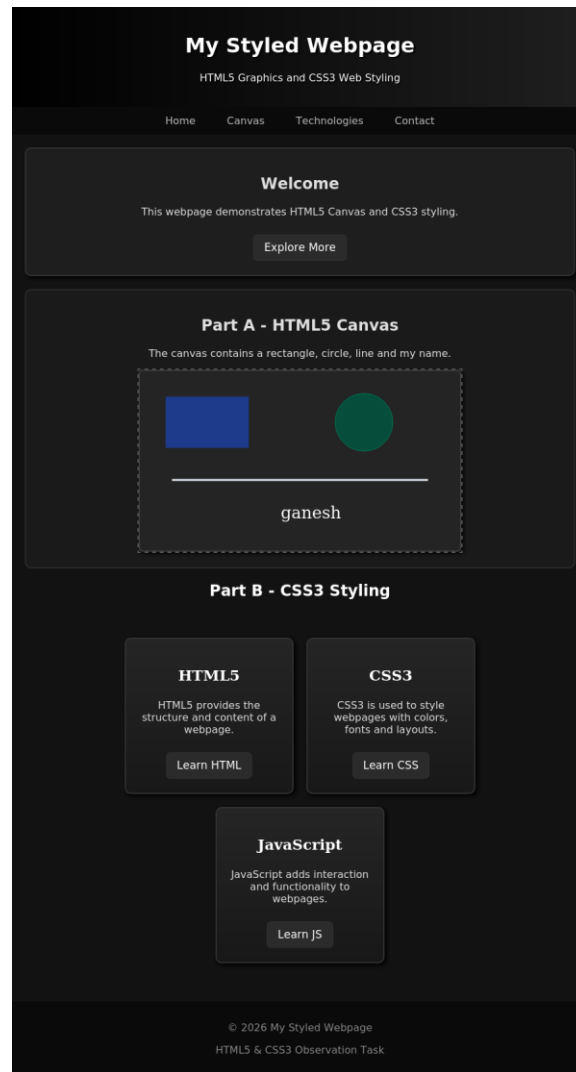
Code	Explanation
<canvas id="myCanvas">	Defines the drawing surface; the width/height attributes set its pixel resolution.
getContext("2d")	Retrieves the 2D rendering context used to issue drawing commands.
fillRect()	Draws a filled, navy-blue rectangle on the canvas.
arc() with fill()/stroke()	Draws a circular path that is filled dark green and outlined in a lighter green.
moveTo() / lineTo()	Defines the start and end points of the horizontal line drawn across the canvas.
fillText()	Renders the student's name as white text on the canvas using a 26px Arial font.
linear-gradient()	Produces the dark tonal gradient backgrounds used on the header and on each card.
box-shadow	Adds layered shadows to the intro box, canvas container, canvas element, and cards.
text-shadow	Adds a subtle shadow behind the header heading text.
border-radius	Rounds the corners of the intro box, canvas, cards, and buttons.
display: flex	Arranges the three technology cards in a centred, wrapping row (.card-container).
:hover	Pseudo-class defining colour, background, and scale changes on mouse-over for nav links and buttons.
<header> <nav> <section> <footer>	Semantic HTML5 elements used to structure the page content.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Load the page in a browser	Page renders with dark background and all sections visible	Page loaded correctly; dark theme displayed	Pass
TC-02	Inspect the canvas graphic	Rectangle, circle, line and name text render inside the canvas	All four elements rendered correctly on canvas	Pass
TC-03	Hover over a navigation link	Link text turns white with a dark grey background	Colour and background changed as expected	Pass
TC-04	Hover over the "Explore More" button	Button background lightens slightly on hover	Background colour changed on hover	Pass
TC-05	Hover over a card button	Button background lightens and slightly enlarges	Button colour lightened; scale transform applied	Pass
TC-06	Resize the browser window (narrow width)	Cards wrap onto new lines via flex-wrap	Cards wrapped correctly below the breakpoint width	Pass
TC-07	Inspect header heading text	Heading displays with a visible drop text-shadow	Text-shadow rendered behind the heading text	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-08	Inspect canvas container corners	Container shows rounded corners and a soft box-shadow	Rounded corners and shadow displayed as styled	Pass
TC-09	Click a navigation anchor link	Page scrolls to the corresponding section id	Browser scrolled to the target section	Pass
TC-10	Inspect footer content	Footer displays copyright text and task label on a dark background	Footer rendered with both lines of text	Pass

9. OUTPUT



10. RESULT

Thus, a styled webpage was successfully designed and developed using HTML5 Canvas and CSS3. The page demonstrates canvas-based graphics rendering, linear-gradient backgrounds, box and text shadows, rounded corners, a responsive Flexbox card layout, and interactive hover effects on navigation links, buttons, and cards. All ten test cases passed successfully, confirming correct rendering and interactive behaviour across the page.